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 SURVEY

A Systematic Literature Review of Hallucinations in Large Language Models

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ABSTRACT This review systematically maps research on hallucinations in large language models using a descriptive scheme that links model outputs to four system architectures: unaugmented generation, post-hoc reactive validation, proactive detection-and-mitigation, and fully integrated detection-and-mitigation designs. Our methodology for this systematic review follows the PRISMA guidelines to ensure transparency and reproducibility. We searched IEEE Xplore, ACM Digital Library, and ScienceDirect for studies published between 2015 and January 2025 and extracted 125 peer-reviewed papers across nine application domains. Quantitative analysis shows that question answering and multimodal tasks account for 48% of all papers, whereas software engineering, educational technology, and autonomous systems are underexplored. Although 87.5% of the studies rely on additional reactive or proactive defenses, only 8.8% implement integrated architecture-level safeguards, revealing a critical gap in unified and dynamic architectures. The resulting classification matrix and domain map provide a diagnostic tool for locating blind spots and comparing architectural maturity. Three actionable priorities emerge: develop integrated reasoning-and-verification loops that pre-empt hallucinations; transfer proven causal-intervention and multi-agent validation pipelines to high-stakes, under-represented domains and benchmark them under real conditions; and build modular, cross-domain evaluation frameworks that isolate the contribution of individual mitigation components and support ablation studies. By consolidating fragmented evidence and quantifying architecture-domain imbalances, this review establishes a traceable foundation for engineering reliable, explainable, and domain-adaptable countermeasures to hallucinations in generative language technology.

INDEX TERMS Large language models, hallucinations, architecture, detection techniques, mitigation strategies, systematic literature review.

I. INTRODUCTION

Large Language Models (LLMs) such as GPT-3, GPT-4, and PaLM have demonstrated remarkable capabilities in natural language understanding and generation [1], [2], question answering [3], code generation [4], powering advancements in dialogue systems [5], [6], and summarization [7], [8]. These models achieve their performance through large-scale pretraining on massive textual corpora, learning statistical patterns that allow them to produce fluent, contextually

appropriate language [9]. However, a growing body of evidence has raised concerns about a fundamental limitation: hallucinations.

Hallucinations are instances where LLMs generate content that is grammatically fluent and syntactically coherent, yet factually incorrect, unverifiable, or inconsistent with a source input [10]. This issue manifests across nearly all natural language generation tasks. In summarization, models may fabricate facts not present in the source document [11]; in question answering [9], they may confidently present incorrect responses [9]; and in scientific or medical domains, they may generate plausible-sounding

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yet fabricated references [2]. Hsu and Roberts [2] found that LLMs used in clinical NLP pipelines can introduce annotations that appear consistent but lack verification, raising concerns about factual integrity. Bzdok et al. [1] similarly observed that LLM outputs often reflect “syntactically elegant guesses,” particularly when models are trained on unstructured or domain-general corpora.

Despite recent innovations, including retrieval-augmented generation [12], Chain-of-Thought prompting [13], role-playing multi-agent prompting [14], and semantic entropy-based detection [10], the hallucination problem remains unresolved. Current models prioritize fluency and likelihood maximization over factual accuracy, and existing evaluation metrics often fail to detect hallucinated content [9], [11]. Moreover, Ji et al. [9] emphasize that mitigation techniques vary greatly in effectiveness across domains and models, and that there is limited consensus in the literature regarding the most effective or generalizable solutions.

Although the phenomenon of hallucination also occurs in humans, the phenomenon of hallucination in LLMs has not yet been systematically investigated from a unified conceptual and methodological lens. Most existing works either address hallucination as a side issue or focus narrowly on task-specific metrics. While several recent works survey hallucination phenomena in LLMs [15], [16], [17], covering aspects such as causes [18], definitions [19], taxonomies [20] and mitigation strategies [21] to varying extents, these efforts remain limited in scope or depth, and none attempt a unified synthesis across these dimensions. For example, Huang et al. [20] provide a comprehensive taxonomy but do not ground their approach in a formal theoretical framework. Reddy et al. [15] and Perković et al. [16] offer useful overviews, yet do not adopt a unifying conceptual framework. Maleki et al. [19], on the other hand, focus on a meta-level analysis of how the term ‘hallucination’ is used across different AI subfields.

This paper addresses this gap through a systematic review of the literature on hallucinations in LLMs, analyzing 125 peer-reviewed publications within a descriptive organizational scheme based on system architectures. The primary objective is to categorize and evaluate existing strategies for detection and mitigation, identifying their applicability and architectural sophistication across various domains such as Question Answering & Information Retrieval, Multimodal & Vision-Language Models, and Biomedical & Scientific Applications. A secondary objective is to examine how this architectural framework provides a useful lens for understanding hallucination behaviors in artificial systems and for future design principles.

We aim to answer the following research questions:

- How can system architecture categories be used to analyze and categorize hallucinations in LLMs?
- In what ways can this architectural framework support the classification of existing hallucination mitigation strategies?

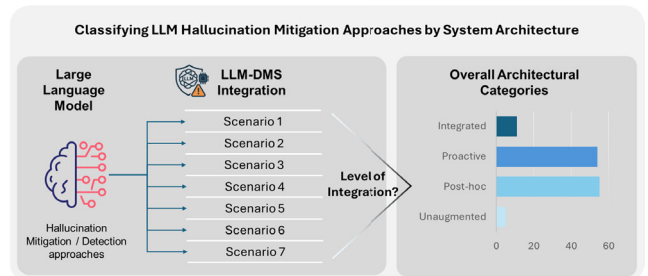


FIGURE 1. 125 papers across 9 domains were analyzed and mapped to system architecture categories; only 8.8% reached Integrated LLM-DMS Architectures.

- Which hallucination detection and mitigation approaches demonstrate distinct architectural sophistication, and what characterizes them as particularly promising?
- Which domains are currently underrepresented in hallucination research, and how does their coverage compare in terms of architectural sophistication?

To address these questions, we performed a domain-specific classification of the hallucination literature and proposed a multiaxial evaluation matrix. We considered papers published between 2015, and January 17, 2025, reporting empirical results, methodological innovations, or theoretical insights related to hallucinations in LLMs. Our methodology follows the PRISMA [22] guidelines to ensure transparency and reproducibility.

By consolidating scattered results and linking them to architectural frameworks, this work provides a solid foundation for future research on robust, explainable, and architecturally sound language models.

To address the identified research gap, this article presents a systematic literature review (SLR) on hallucinations in LLMs, combining empirical categorization with an architecturally grounded analytical lens (Figure 1). Drawing from our proposed system architecture categories, 125 peer-reviewed studies are evaluated along two key axes: their domain of application and the sophistication of their hallucination detection and mitigation strategies. This structured synthesis provides a comprehensive overview of the current research landscape, identifying dominant areas of focus, emerging theoretical advancements, and significant gaps. The proposed classification matrix serves both as a conceptual framework and a diagnostic tool to support the development of more explainable, architecturally aligned, and domain-adaptable solutions to hallucination phenomena in generative AI systems.

With regard to the research gap, the main contributions of this work are as follows:

- Introduction of a novel descriptive organizational scheme based on system architectures, enabling a structured analysis of hallucination detection and mitigation strategies in LLMs across diverse domains.
- Systematic mapping of 125 peer-reviewed publications, revealing a concentration of research in domains such as Question Answering & Information Retrieval and

Multimodal & Visual-Language Models, while identifying substantial gaps in Software Engineering & Code Generation, Educational AI & Learning Systems, and Autonomous Systems & Robotics.

- Evaluation of architectural sophistication across reviewed techniques, highlighting that only 8.8% of studies exhibit Integrated LLM-Detection and Mitigation System (DMS) Architectures, indicating the need for more dynamic and unified system designs in future research.

This work is structured as follows: In section II, preliminary concepts and fundamentals are elaborated to give a brief introduction to the topic. Next, section III, describes our SLR methodology. The results from our SLR are then presented in section IV and consequently discussed in section V. Finally, we conclude in section VI by highlighting the theoretical and practical implications of our findings.

II. PRELIMINARY CONCEPTS AND FUNDAMENTALS

Hallucination in LLMs refers to the generation of content that is fluent, syntactically correct, and semantically plausible, but factually incorrect, unfounded, or completely fabricated. They differ from conventional natural language generation errors by a high degree of linguistic coherence that can mask underlying factual inaccuracies [9], [10]. Ji et al. [9] define hallucination as inaccurate generation and propose a multidimensional taxonomy based on origin (intrinsic vs. extrinsic), nature (factual, logical, linguistic), and manifestation (addition, omission, distortion).

Evaluating hallucinations remains difficult. Classical metrics like ROUGE or BERTScore fail to capture the factual accuracy [11]. More recent efforts include entailment-based metrics (e.g., FactCC), QA-based approaches (e.g., QAGS), and semantic methods such as SummaC and semantic entropy [10], [23], [24]. Yet these techniques vary in their reliability across domains, and human evaluation remains the gold standard despite cost and subjectivity. Hallucination also manifests differently across tasks: abstractive summarization systems may generate unsupported facts; dialogue models can produce fictional information when context is missing; and data-to-text systems frequently hallucinate content not present in the structured input [9], [11].

Mitigation efforts aim to reduce hallucinations through architectural or inference-level innovations. Retrieval-augmented generation (RAG) incorporates external factual knowledge to support grounded outputs [12], while Chain-of-Thought prompting encourages intermediate reasoning steps [13]. Despite these advances, no approach has fully eliminated hallucinations in open-ended text generation tasks.

In addition to fully systematic reviews, the broader literature also contains relevant but less formally structured analyses. While Table 1 presents structured and methodologically grounded literature reviews focused specifically on hallucinations in LLMs, several additional articles address

related topics but are not considered systematic reviews. These works do not follow a standardized review methodology, such as PRISMA or domain-specific taxonomies, and typically do not provide transparent inclusion criteria or structured synthesis of reviewed studies. However, they offer valuable information about the broader discourse around hallucinations and limitations in LLMs. Zhang et al. [21] provide a comprehensive study of safety, security, and privacy issues in LLMs, including hallucinations, and propose a refined classification framework to distinguish these concerns and their corresponding defenses. Sanu et al. [18] present a general critique of LLM vulnerabilities, including hallucinations, bias, adversarial attacks, and outdatedness, framed as ‘loopholes’ in their deployment. Reddy et al. [15] offer a conceptual discussion on the nature and types of hallucinations in LLMs, supported by real-world examples but lacking empirical aggregation of research. Perković et al. [16] focus on the technical and social dimensions of hallucinations, such as data provenance and trust, while proposing mitigation strategies from a systems engineering perspective. Lastly, Huang et al. [20] discuss hallucinations within a broader overview of generative AI’s limitations, offering definitions and conceptual categories rather than systematically analyzing existing work.

Although not meeting the formal review criteria, these articles contribute important perspectives and help contextualize the growing interest in LLM hallucination research.

A. EVOLUTION OF HALLUCINATIONS IN LARGE LANGUAGE MODELS

The problem of hallucination has evolved in parallel with the growing capabilities of LLMs. Early neural language models such as GPT-2 and BART showed signs of hallucination, particularly in abstract summarization [9], [11]. Originally, these problems were attributed to overfitting or weak attentional mechanisms [9], [11]. However, with the advent of transformer-based models such as GPT-3, GPT-4 and PaLM, hallucinations became clearer and better understood as a systemic result of predicting the next token based on noisy, incomplete, or unstructured training data [9], [25].

LLMs have also been shown to hallucinate in specialized domains such as clinical NLP and neuroscience. Hsu and Roberts [2] demonstrated that LLMs used in weak supervision pipelines for clinical named entity recognition can introduce semantically plausible but unverified labels, raising concerns about label reliability. Similarly, Bzdok et al. [1] highlighted that LLMs may produce plausible but ungrounded outputs, especially when trained on unstructured or domain-unspecific data, contributing to the broader understanding of hallucinations as coherent yet misaligned model behaviors. Farquhar et al. [10] proposed the semantic entropy framework to detect such cases and showed that high variance in semantic intent across resampled outputs correlates with hallucination risk.

TABLE 1. Overview of recent literature reviews regarding hallucinations in LLMs. The table displays each review's authors, publication year, and the number of papers included in the review. Additionally, a summary is given, and the applied review structure is provided.

Authors	Year	# of papers included	Summary	Structure applied to categorize literature
Lavrinovics et al. [17]	2025	Not specified	Examines the mitigation of hallucinations in LLMs through the integration of knowledge graphs (KGs). It categorizes hallucination types, compares evaluation resources, and proposes architectural stages for knowledge integration - pretraining, inference, and post-generation. The paper emphasizes the limitations of current evaluation methods, highlights the importance of multilingualism, and calls for parameter-efficient mitigation strategies.	(1) Hallucination Types: Intrinsic vs. Extrinsic (2) Evaluation Resources: Categorization of benchmark datasets and evaluation methods (3) Knowledge Integration Stages: Pretraining, Inference, and Post-generation (4) Model Architecture Focus: Structural integration of Knowledge Graphs (KGs) across the LLM lifecycle (5) Application Dimensions: Consideration of multilingual and domain-specific settings
Maleki et al. [19]	2024	333	A comprehensive and cross-domain systematic review of the term "AI hallucination" is presented, analyzing 333 scientific papers from 14 large databases from different fields such as computer science, healthcare, law, and ethics. The paper critically examines the inconsistency in the definitions and usage of the term "hallucination" in the context of LLMs, revealing its metaphorical ambiguity and domain-dependent interpretation. By categorizing the definitions according to domain of application and highlighting alternative terms (e.g., fabrication, stochastic parroting, confabulation), the study underlines the need for terminological clarity and semantic precision. It concludes with a call for standardized definitions and proposes a taxonomy to better capture the nuanced nature of factual errors in generative AI systems.	(1) Definition Sources: Extracted from 14 databases across domains (e.g., health, law, technology) (2) Application Domains: Grouping by usage context: Chatbots, Dialogue Systems, Healthcare, Legal/Ethical, Academia, etc. (3) Linguistic Framing: Analysis of alternative terminologies (e.g., fabrication, stochastic parroting, confabulation) (4) Semantic Characteristics: Emphasis on factuality, plausibility, coherence, and deviation from established knowledge (5) Temporal Framing: Evolution of definitions pre- and post-ChatGPT (i.e., before and after November 2022)

As awareness of the issue grew, the research community shifted from post-hoc detection to active mitigation. Retrieval-augmented models like RAG introduced dynamic grounding by retrieving relevant factual data during inference [12]. Chain-of-Thought prompting showed promise in mathematical and commonsense reasoning tasks by mimicking deliberative thought processes [13]. Reinforcement learning from human feedback (RLHF) also became a prominent method for aligning outputs with user expectations and reducing hallucination frequency [26].

Recent work has further expanded the scope of hallucination research into new domains. Zhang et al. [25] proposed an Entity-Relation-based reinforcement learning framework to reduce spatial hallucination in path planning, highlighting that the issue extends beyond text to reasoning in structured environments. Despite these advances, hallucination remains unsolved, not only in open-domain generation, where prompts are vague and training data may be contradictory, but also in structured tasks, as recent research has shown. The evolution of hallucination research reflects an ongoing tension between scale, fluency, and factual grounding.

III. METHODOLOGY

A. OVERVIEW

This section outlines the methodological steps adopted for this study, which comprises two primary components: (i) a

systematic literature review on hallucinations in LLMs, and (ii) a categorization framework for synthesizing the identified papers. The former describes how relevant studies were identified, screened, and included, while the latter discusses the approach for classifying the selected studies in terms of domain focus and architecture categories.

B. SYSTEMATIC LITERATURE REVIEW PROCEDURE

We conducted a SLR following the preferred reporting items for systematic reviews and meta-analyses (PRISMA) guidelines by Page et al. [22]. An SLR systematically gathers, evaluates, and synthesizes relevant evidence on a defined topic to establish the state of the art, identify research gaps, and provide a solid foundation for further investigation. In our case, the SLR was designed to address the research questions outlined in the Introduction, with a focus on hallucination phenomena in LLMs, evaluated through a domain-based and an organisational-technical categorization. To ensure a structured and transparent review, we followed eight methodological steps adapted from the PRISMA guidelines [22]. These steps are summarized in Figure 2 and described in more detail below.

1) LITERATURE SEARCH STRATEGY

Following the presented framework, the next step of this SLR was to define the search string and the criteria to exclude

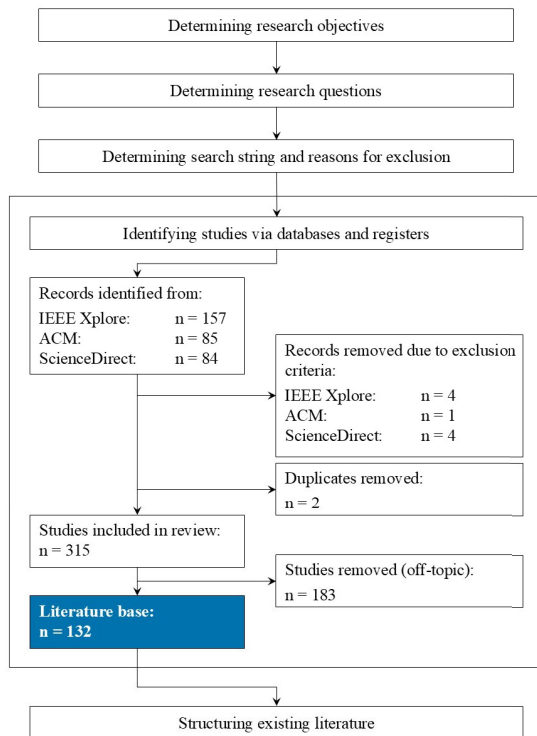


FIGURE 2. Overview of the process for obtaining the literature base as derived from the PRISMA guidelines [22].

individual search results. Our search string consists of the following components:

Title, Abstract & Keywords: (“hallucination*”) AND (“LLM*” OR “large language model*”)

As the research goal of this article is to investigate hallucinations in LLMs, abbreviated LLMs, the search term was formulated to comprise three interrelated components: hallucinations, large language models, and their variations. To achieve this, the terms “hallucination*”, “LLM*”, and “large language model*” were included in the search string. Boolean operators were used to ensure accuracy and completeness. The “AND” operator was used to find only those papers that address all key dimensions simultaneously, while the “OR” operator expanded the scope by including synonymous terms and variations. In addition, the use of the wildcard (*) enabled the inclusion of plural forms and variations of the terms, further improving coverage.

To ensure the quality of the identified work, the following exclusion criteria were defined:

Exclusion Criteria

- The publication was published before 2015.
- The publication type is not a peer-reviewed conference paper or journal article. To ensure academic rigor, formats such as poster sessions, editorials, interviews, commentaries, and research-in-progress papers are excluded.

- The publication is considered “off-topic” by at least two authors. This criterion ensures the relevance of the included literature.
- The publication is written in a language other than English.

In the fourth step, the databases used for the SLR were selected and searched. The present study used three databases: IEEE Xplore Digital Library, ACM Digital Library, and ScienceDirect database. The search term combination remained the same for all databases and was applied to search titles, abstracts, and keywords. IEEE Xplore was last accessed on 01/17/2025, ACM on 01/17/2025, and ScienceDirect on 01/17/2025.

2) STUDY SELECTION AND EXTRACTION

Before screening, four articles from IEEE Xplore, one article from ACM, and four articles from ScienceDirect were excluded due to incorrect article types. Two other articles were removed because they were duplicates.

Thus, a total of 315 articles were screened. In the screening process, the authors examined the title and the abstract for the content and semantic relationship to the research question of this paper. To avoid bias in the selection process, all identified papers were individually reviewed by two reviewers.

First, all papers were rechecked for terms contained in the search string. After this first check, the title and abstract were checked using the exclusion criteria controlling the semantic context of the respective work. These papers passing this check were downloaded and stored for information extraction.

After review by the authors, the reliability of the results was 98.41% (agreement on 310 of 315 total articles). No automation tools were used for the search.

As part of the screening process, several articles were excluded to ensure the relevance and alignment of the selected literature with the objectives of this study. A total of 183 articles were deemed not relevant to this work for the following reasons:

- **Superficial or casual references to hallucinations:** Many articles mentioned hallucinations only briefly as a general risk, theoretical problem, or casual observation without providing empirical analysis, proposing new detection or remediation methods, or providing insights into hallucination-specific behavior in LLMs.
- **Application without analytical depth:** A considerable proportion of the excluded papers implemented LLMs for domain-specific applications without analyzing or addressing hallucinatory behavior. These studies often used hallucinations as a motivation, but did not directly investigate the phenomenon or propose solutions.
- **Lack of methodological or conceptual contribution:** In order to maintain the scientific rigor of the analysis, publications were excluded if they offered little that was new (e.g., repetition of known results), contained no empirical evaluation, or showed minimal theoretical or methodological development.

- **Focus on neighboring but different concerns:** Several papers addressed topics such as AI bias, trustworthiness, explainability, user acceptance, or ethical use of generative models. These studies were excluded if hallucinations were not a central and analytically investigated component of the work.

In total, 132 articles were included in the preliminary work and used for the SLR (IEEE Xplore (n = 76), ACM (n = 44), ScienceDirect (n = 12)).

C. CLASSIFYING THE RESULTS WITH A DESCRIPTIVE ORGANIZATIONAL SCHEME FOR LLM OUTPUTS

Of the 125 studies identified, 7 were categorized as literature reviews and excluded, we implemented a structured classification scheme to better understand the landscape of research on LLM hallucinations. To this extent, we developed and employed a descriptive organizational scheme designed to be functionally precise and engineering-oriented. This approach is essential for analyzing system-level failures and mitigation strategies, as it provides a taxonomy that maps directly to system architecture and observable behaviors. To this end, we developed and employed a descriptive organizational scheme to categorize the phenomena reported in the selected literature.

This scheme is derived from a synthesis of current research on AI-generated hallucinations and their mitigation strategies [9]. It organizes system architecture components and their outputs into two primary categories based on architectural design: (A) the output of an unaugmented LLM operating on its internal parametric knowledge, and (B) the output of an LLM augmented with an external DMS. A prevalent example of a DMS in the literature is the RAG architecture [12]. This architectural distinction is fundamental, as the introduction of a DMS creates a new set of potential success and failure modes that must be analyzed independently [25]. The scheme outcome details seven mutually exclusive scenarios, which are presented in Tables 2 through 8 and will be used to classify the contributions of the reviewed papers.

1) SYSTEM CHARACTERISTICS WITHIN THE ORGANIZATIONAL SCHEME

To understand the seven scenarios of output generation, it is first necessary to define the fundamental characteristics of the two primary system architectures identified in our scheme: The unaugmented LLM and the LLM augmented with a DMS.

Unaugmented LLM Characteristics: An unaugmented LLM generates output based solely on its internal, parametric knowledge learned during training. This architecture is defined by the following characteristics [9], [12]:

- **Reliance on parametric memory:** The model's responses are generated directly from the knowledge stored in the parameters of the neural network. The behavior of the model is a direct function of the

training goal, which is usually the prediction of the next token [12], [26].

- **Static knowledge base:** The internal knowledge of the model is fixed after the training phase, which leads to a "knowledge cutoff". It cannot access real-time information or easily update its knowledge base without being retrained or fine-tuned [9].
- **Vulnerability to artifacts in the training data:** The model is prone to generating hallucinations due to noise or bias in the rich training data. The default training target does not explicitly reward fidelity to a particular source document, which can lead to unfaithful or nonsensical results [9], [11], [26].
- **Single-stage generation:** The output is usually generated in a single, continuous forward pass, which makes the process fast but relatively opaque, as there are no intermediate components that need to be verified for their origin [12].

Augmented LLM Characteristics: An augmented system integrates the core LLM with one or more external components designed to detect and mitigate hallucinations. This architecture has a different set of operational characteristics:

- **Multi-Stage Processing:** These systems operate in multiple stages, such as retrieval-then-generation (as in RAG) or generation-then-correction. This architecture allows for explicit points of intervention and verification [9], [12].
- **Access to Non-Parametric Memory:** Many DMS architectures, particularly RAG, augment the LLM by providing it with access to an external, non-parametric knowledge base (e.g., a vector index of Wikipedia). This allows the model to ground its output in up-to-date, verifiable information [12].
- **Explicit Verification Mechanisms:** The DMS component is designed to perform fact-checking or consistency-checking. This can involve comparing generated text against retrieved documents using Natural Language Inference (NLI) models or Question Answering (QA) systems [9].
- **Improved Interpretability and Controllability:** By separating knowledge retrieval from generation, these systems can provide better interpretability. The retrieved documents serve as proof of origin for the generated output and explain why the model generated a particular response [12].
- **New Potential Failure Points:** The added complexity introduces new vulnerabilities. The system can fail if the DMS retrieves irrelevant or flawed information, or if the detection logic itself is unable to identify subtle inconsistencies, a significant focus of current research [9], [23].

2) SCENARIOS FOR UNAUGMENTED LLM OUTPUTS

The first two scenarios are the outcome of the baseline behavior of a standalone LLM, as conceptually illustrated in

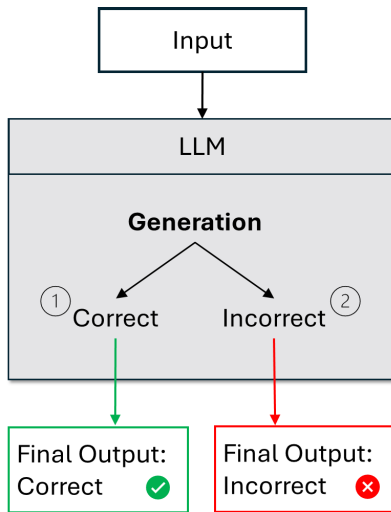


FIGURE 3. Conceptual illustration of an unaugmented LLM generation architecture.

Figure 3. Its outputs are a direct function of its training and internal knowledge [26].

- **Scenario 1: Unassisted Correct Generation** represents an ideal case where the model’s probabilistic generation aligns with factual reality.
- **Scenario 2: Unchecked Hallucination** is the canonical failure mode that has motivated a vast body of research, an issue that has been thoroughly documented in recent surveys [9]. Hallucinations are formally defined as generated content that is nonsensical or unfaithful to the provided source content [9], [10]. They are known to arise from several sources: flaws in the training data can cause the model to learn and store factually incorrect information or “wrong correlations” within its parameters; they can stem from parametric knowledge gaps where the model’s knowledge is outdated or incomplete (i.e., knowledge cutoffs); or they can be the result of flawed decoding strategies during the generation process itself [9]. This scenario represents the raw, unmitigated output of a system lacking any corrective mechanisms.

3) SCENARIOS FOR DMS AUGMENTED LLM OUTPUTS

The other five scenarios encompass systems where a second component is integrated to verify and/or correct the LLM’s output, as conceptually illustrated in Figure 4. This introduces a multi-stage process where new outcomes are possible.

- **Scenario 3 (Verified Correct Generation)** and **Scenario 4 (Successful Hallucination Correction)** represent the intended successes of an augmented architecture. In Scenario 4, DMS effectively converts a system failure into a success, demonstrating the primary value proposition of architectures like RAG, which aim to ground LLM outputs in verifiable, retrieved evidence, thus reducing hallucination [12].

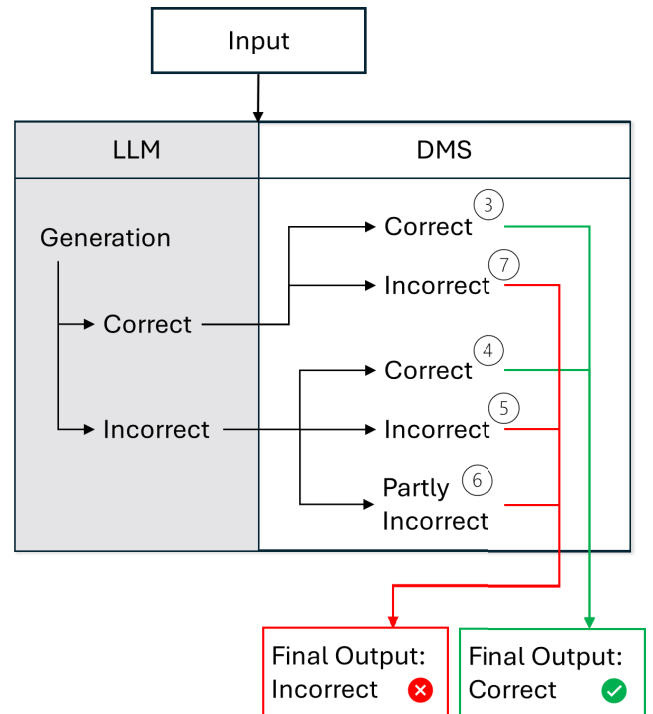


FIGURE 4. Conceptual illustration of a DMS augmented LLM architecture.

- The remaining scenarios delineate distinct and critical failure modes within these more complex augmented systems. **Scenario 5 (Unrecognized Hallucination)** occurs when the DMS fails to detect the initial error. [25]. This is a significant focus in the literature, often attributed to specific failure points within the DMS itself, such as the retrieval of irrelevant or misleading documents, or the failure of the “detection” mechanism to identify subtle inconsistencies [9], [23], [24].
- **Scenario 6 (Partially Corrected Hallucination)** describes a gradual failure where the DMS addresses some, but not all, inaccuracies in a complex output. This can result from the failure of the LLM to fully synthesize multiple pieces of retrieved context or from incomplete evidence retrieval by the DMS [9].
- Finally, **Scenario 7 (Corrupted by Correction)** captures a paradoxical and highly problematic failure in which the DMS itself introduces an error into an otherwise correct LLM output. This can occur if the retrieved knowledge is flawed or if the DMS’s corrective logic is too aggressive, corrupting a nuanced and accurate initial response [9]. This highlights that the mitigation layer itself is a potential source of error that requires independent validation.

D. ANALYSIS AND CATEGORIZATION FRAMEWORK

To systematically evaluate and categorize existing research on hallucinations in LLMs, this study introduces an analytical framework based on system architecture. The objective is

to assess each research paper by examining how effectively its proposed architecture integrates the LLM with the DMS. Additionally, this framework explicitly considers the seven scenarios, enhancing theoretical rigor and practical applicability.

This framework is structured along two key dimensions:

System Architecture Categories: This dimension categorizes all identified systems along four distinct architectural archetypes. Each category is directly linked to the identified scenarios.

- **Unaugmented LLM Generation Architectures:** This category represents systems where the LLM operates autonomously, without any structured oversight or integrated DMS. Consequently, LLM outputs depend exclusively on internal parametric knowledge acquired during training. Outputs can either be factually correct (scenario 1) or incorrect due to inherent limitations, such as training data biases or spontaneous hallucinations (scenario 2).
- **Post-hoc Reactive Validation Architectures:** This category encompasses systems where a standalone DMS is coupled with the LLM but operates exclusively in a reactive manner. The DMS is triggered only after the LLM has completed its generation, typically based on predefined heuristics, confidence thresholds, or anomaly signals. As a result, this architecture allows for a range of outcomes, including Scenario 3–7, depending on whether and how the DMS intervenes. Corrections, if applied, are handled entirely outside the LLM, which remains a static, unrefined generator. Evaluation criteria for this architecture include the precision and recall of post-hoc detection mechanisms.
- **Proactive DMS Architectures:** This category is defined by a proactive and tightly coupled interaction between the LLM and the DMS. Unlike reactive systems, the DMS in this architecture anticipates potential errors and actively guides or constrains the generation process before and during output production. This enables outcomes aligned with Scenarios 3–7, as each output undergoes structured verification or correction in real time. Scenarios 1 and 2 are inherently excluded, since no output is provided without undergoing some form of proactive validation. Evaluation criteria for this architecture focus on the effectiveness of preemptive error detection mechanisms and the degree to which they suppress complex failure modes, particularly Scenarios 5–7. These systems often incorporate bidirectional feedback loops, enabling dynamic refinement of the LLM's generation pathway based on real-time assessments of factual accuracy or coherence.
- **Integrated LLM-DMS Architectures:** This category is characterized by architectures that move beyond the conventional separation of LLM and DMS components. Instead, the system fully integrates both components into a unified, dynamic, and proactive framework to

minimize hallucinations. The DMS not only anticipates errors before but also operates through internal feedback loops, meta-reasoning, and self-validation.

2. Domain: Classification of research based on the area in which their approach is applied. This helps identify areas that are most advanced in the mitigation of hallucinations and serves as a basis for subsequent analysis. The following domains have been identified:

- **Software Engineering & Code Generation:** Involves models generating or assisting with source code, highlighting precision and accuracy in algorithmic logic and syntax.
- **Question Answering & Information Retrieval:** Covers methods that retrieve and generate accurate and relevant information in response to user queries, including RAG pipelines and grounding mechanisms to reduce factual errors.
- **Ethical & Trustworthy AI:** Focuses on developing systems that behave ethically, transparently, and reliably, addressing hallucinations through mechanisms like toxicity detection, fact-checking, or content verification to ensure safe deployment.
- **Multimodal & Vision-Language Models:** Covers models that integrate visual, textual, or multimodal data (e.g., image captioning, spatial reasoning, video-language alignment), where hallucinations can arise from cross-modal misalignment or grounding failures.
- **Domain-Specific QA (e.g., Manufacturing, Legal AI):** Targets highly specialized domains (e.g., manufacturing, legal, financial, or risk assessment), where factual precision is critical and hallucinations must be addressed using domain-specific knowledge sources and constraints.
- **Biomedical & Scientific Applications:** Covers sensitive, high-stakes fields (e.g., medicine, biology, scientific research) where factual errors can have critical consequences, and hallucination mitigation often involves rigorous domain-specific verification and citation strategies.
- **User-Centric AI & Explainability:** Focuses on enhancing user trust through explainable, transparent outputs, often using UI features, feedback loops, or user-guided validation mechanisms to help users detect or correct hallucinations.
- **Autonomous Systems & Robotics:** Refers to real-time decision-making scenarios in robots and autonomous platforms, demanding consistently reliable outputs.
- **Educational AI & Learning Systems:** Involves personalized learning and knowledge assessment systems, emphasizing accuracy to ensure effective educational outcomes.

IV. RESULTS

The SLR initially identified 132 publications related to hallucinations in LLMs. Out of these, 125 were included in the final analysis, while the remaining 7 were classified

TABLE 2. Unassisted correct generation. An LLM outputs a factually correct response using only its internal knowledge, without external verification.

Unaugmented LLM Unassisted Correct Generation (Scenario 1)	
Process	Input → LLM processes → Final Output: Correct
Interpretation	Represents the ideal outcome of an unaugmented LLM. Relying solely on internal parametric knowledge, the model produces a factually accurate response. This correctness isn't from deliberate reasoning but emerges when learned statistical patterns align with reality. While showcasing the model's capabilities, the outcome remains unreliable without external verification.
Final Output	Correct

TABLE 3. Unchecked hallucination. A LLM produces a confident but incorrect output, with no mechanism to detect or prevent the error.

Unaugmented LLM Unchecked Hallucination (Scenario 2)	
Process	Input → LLM processes → Final Output: Incorrect
Interpretation	This is the canonical failure mode of an LLM: it generates a fluent, confident, yet factually wrong response. With no DMS present, the hallucination reaches the user unchecked. This poses the greatest risk of deploying unaugmented LLMs, as the system can't distinguish success from failure. Root causes include flawed training data, knowledge gaps, or misinterpreted user intent.
Final Output	Incorrect

TABLE 4. Verified CORRECT GEneration. The DMS confirms the LLM's correct output, ensuring reliability and trust.

Augmented LLM Verified Correct Generation (Scenario 3)	
Process	Input → LLM processes (Correct) → DMS Confirms → Final Output: Correct
Interpretation	In this scenario, the LLM produces a correct response, and the DMS successfully verifies it. Acting as a validation layer, the DMS turns an unverified claim into a confirmed fact (e.g., via RAG), providing a "seal of approval" that builds trust—especially in high-stakes applications.
Final Output	Correct

TABLE 5. Successful hallucination correction. The DMS identifies and corrects a faulty LLM output (e.g., via retrieval), highlighting the strength of augmented systems.

Augmented LLM Successful Hallucination Correction (Scenario 4)	
Process	Input → LLM processes (Incorrect) → DMS Corrects → Final Output: Correct
Interpretation	This is the quintessential success case for a DMS, converting system failure into success. The DMS detects the LLM's hallucination and applies a mitigation strategy, typically retrieving correct information and guiding the LLM to generate a fact-based answer. This demonstrates the core value of augmented architectures in improving system reliability.
Final Output	Correct

TABLE 6. Unrecognized hallucination. The LLM generates an incorrect output, and the DMS fails to intervene, creating a false sense of reliability.

Augmented LLM Unrecognized Hallucination (Scenario 5)	
Process	Input → LLM processes (Incorrect) → DMS Fails to Detect → Final Output: Incorrect
Interpretation	A critical failure mode where the LLM hallucinates and the DMS fails to detect it. Arguably more dangerous than an unchecked hallucination, it creates a false sense of security, as users trust outputs from a "safety-checked" system. Such failures often stem from DMS limitations, like retrieval errors (missing or poorly ranked context) or weak detection mechanisms.
Final Output	Incorrect

TABLE 7. Partially corrected hallucination. The DMS partially corrects a flawed output, yielding a seemingly reliable response that still harbors factual errors, posing a deceptive risk.

Augmented LLM Partially Corrected Hallucination (Scenario 6)	
Process	Input → LLM processes (Incorrect) → DMS Partially Corrects → Final Output: Partially Incorrect
Interpretation	This gradual failure occurs when an output contains multiple inaccuracies, and the DMS corrects some but not all of them. The final output is an improvement but remains factually flawed. This can happen due to incomplete context retrieval or the LLM failing to incorporate all corrective facts from the provided context. Because the output is mostly correct, users may be more inclined to trust the remaining flawed details.
Final Output	Partially Incorrect

TABLE 8. Corrupted by correction. The LLM generates a factually correct output, but the DMS unintentionally introduces an error, corrupting the original response.

Augmented LLM Corrupted by Correction (Scenario 7)	
Process	Input → LLM processes (Correct) → DMS Introduces Error → Final Output: Incorrect
Interpretation	A paradoxical failure where the safety system itself introduces an error. The LLM produces a correct response, but the DMS corrupts it. This can be caused by the DMS relying on flawed external knowledge (e.g., an incorrect RAG document), over-correcting a nuanced statement, or misinterpreting the user's intent. This scenario is profoundly damaging to trust and highlights the need to validate the DMS itself, not just the LLM.
Final Output	Incorrect

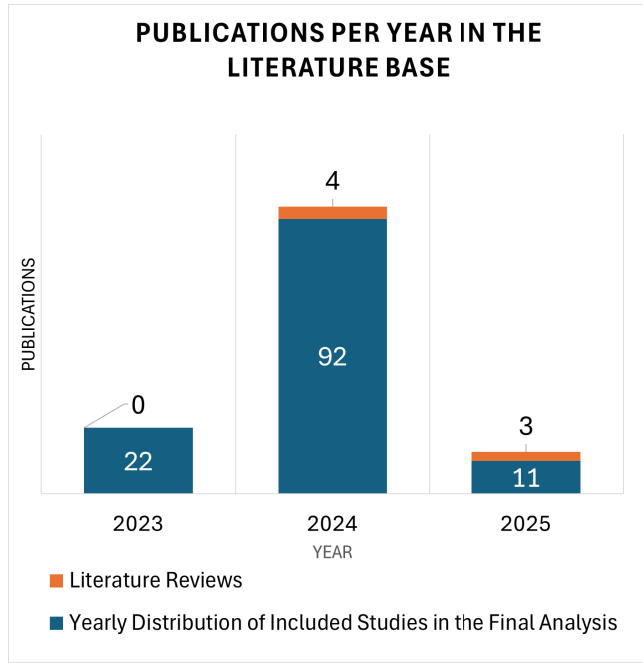


FIGURE 5. Yearly distribution of publications included in the analysis (blue), with excluded literature reviews shown for reference (orange). Only papers published between 2015 and January 17, 2025, were included.

as literature review papers. These reviews, although relevant for contextual understanding, were excluded from further categorization and analysis to maintain focus on empirical and technical contributions.

The bar chart in Figure 5 presents the distribution of the included studies by year. A clear increase in research interest is observable in 2024, which accounts for the vast majority of contributions (92 out of 125), while 2023 and 2025 contributed 22 and 11 studies, respectively. The sharp rise in 2024 aligns with the growing adoption and scrutiny of generative LLMs. The literature reviews were distributed across 2024 (4) and 2025 (3), with none in 2023.

A. CATEGORIZATION BY DOMAIN AND SYSTEM ARCHITECTURE

The classification framework used for this review includes two main axes:

The classification matrix in Table 9 displays the intersection of these two dimensions across all 125 included papers. This matrix serves as the foundation for the following domain- and system architecture.

B. DOMAIN DISTRIBUTION OF RESEARCH

Figure 6 visualizes the distribution of the research papers across the nine identified domains. The most prevalent domain is Question Answering & Information Retrieval, with 32 publications, followed by Multimodal & Vision-Language Models (28), and Domain-Specific QA (14). Conversely, domains such as Autonomous Systems & Robotics (6) and Educational AI & Learning Systems (5) are currently underrepresented in the literature.

This distribution indicates that hallucinations in LLMs are predominantly studied in high-impact application areas such as QA and vision-language tasks.

C. DISTRIBUTION BY ARCHITECTURE CATEGORY

The papers were also analyzed according to their alignment with the four architecture categories. Figure 7 shows the distribution:

- Unaugmented LLM Generation Architectures: 5 papers (4.0%)
- Post-hoc Reactive Validation Architectures: 55 papers (44.0%)
- Proactive DMS Architectures: 54 papers (43.2%)
- Integrated LLM-DMS Architectures: 11 papers (8.8%)

The majority of papers fall under the Post-hoc Reactive Validation Architectures and Proactive DMS Architectures categories, indicating that while many researchers explicitly incorporate reasoning-based frameworks, few push the boundary towards advanced, fully integrated systems.

D. RESULTS BY SYSTEM ARCHITECTURE CATEGORY

1) HIGHLIGHTING UNAUGMENTED LLM GENERATION ARCHITECTURES

This first category, Unaugmented LLM Generation Architectures, is foundational to hallucination research as it includes systems where the LLM operates autonomously, relying exclusively on its internal parametric knowledge without any structured oversight or integrated DMS, as illustrated in Figure 8. Research in this area is critical for quantifying the baseline performance and inherent limitations of LLMs, thereby establishing the scope of the hallucination problem that more advanced architectures seek to solve.

The studies in this category analyze the raw, uncorrected outputs of LLMs to understand their factual accuracy and biases. For instance, Zuccon et al. [34] directly evaluated ChatGPT’s ability to attribute its answers to credible sources

TABLE 9. Overview of the publications in the literature base. The table groups papers by their domain of application and categorizes them according to the architecture exhibited in their hallucination detection and mitigation strategy. In the Domain column (1st X-axis), research is classified into categories such as Software Engineering & Code Generation, Question Answering & Information Retrieval, Multimodal Models, and others. In the architecture category columns (2nd - 5th X-axis), each publication is placed into one of the four previously established categories: Unaugmented LLM Generation Architectures, Post-hoc Reactive Validation Architectures, Proactive DMS Architectures, and Integrated LLM-DMS Architectures. Each cell lists the authors whose work falls under the respective domain and architecture classification.

Domains	Unaugmented LLM Generation Architectures	Post-hoc Reactive Validation Architectures	Proactive DMS Architectures	Integrated LLM-DMS Architectures
Software Engineering & Code Generation		Yu et al. [27]	Alshahwan et al. [28], Cai et al. [29], Quddus et al. [30], Samanta et al. [31], Tanzil et al. [32]	
Question Answering & Information Retrieval	Moayeri et al. [33] Zuccon et al. [34]	Chen et al. [35], Dutta et al. [36], Furumai et al. [37], Hu et al. [38], Huo et al. [39], Koc et al. [40], Li et al. [41], Li et al. [42], Ngu et al. [43], Pezeshkpour et al. [44], Snyder et al. [45], Wei et al. [46], Xia et al. [47]	Alshammary et al. [48], Chen et al. [49], Feng et al. [50], Hang et al. [51], Hu et al. [52], Huang et al. [53], Huang et al. [54], Shiri et al. [55], Song et al. [56], Su et al. [57], Yang et al. [58], Zhang et al. [59], Zhang et al. [60], Zhu et al. [61]	Shi et al. [62], Sun et al. [63], Wang et al. [64]
Ethical & Trustworthy AI	Kraft et al. [65]	Bahrami et al. [66], Ding et al. [67], Hannigan et al. [68], Huang et al. [69], Jiang et al. [70], Li et al. [71], Shah et al. [72], Tripathi et al. [73], Wang et al. [74]	Perrina et al. [75], Yu et al. [76]	
Multimodal & Vision-Language Models	Jiang et al. [77], Kaul et al. [78]	Bendeck et al. [79], Boenisch et al. [80], Ding et al. [81], Halperin et al. [82], Hamid et al. [83], Jiang et al. [84], Kumar et al. [85], Li et al. [86], Ma et al. [87], Verma et al. [88], Wang et al. [89], Yong et al. [90], Yu et al. [91]	Fei et al. [92], Fei et al. [93], Gao et al. [94], Huang et al. [95], Ranasinghe et al. [96], Sarfati et al. [97], Wang et al. [98], Wang et al. [99], Wei et al. [100], Yu et al. [101]	Gao et al. [102], Xu et al. [103], Zhang et al. [104]
Domain-Specific QA		Kang et al. [105], Leung et al. [106], Ruangchutiphophan et al. [107], Vidyaratne et al. [108], Yang et al. [109], Yao et al. [110]	Bei et al. [111], Roychowdhury et al. [112], Sheng et al. [113], Shetty et al. [114], Wang et al. [115], Xu et al. [116], Zheng et al. [117], Zheng et al. [118]	
Biomedical & Scientific Applications		Zhan et al. [119]	Chang et al. [120], Craig et al. [121], Herron et al. [122], Pang et al. [123], Tariq et al. [124], Zeng et al. [125], Zhao et al. [126]	Sharma et al. [127], Xu et al. [128]
User-Centric AI & Explainability		Leiser et al. [129], Leiser et al. [130], Li et al. [131], Maragheh et al. [132], Procko et al. [133], Waldo et al. [134], Zhao et al. [135]	Cheng et al. [136], Lin et al. [137], Zhao et al. [138]	Sun et al. [139], Zhang et al. [140]
Autonomous Systems & Robotics		Li et al. [141], Li et al. [142], Manas et al. [143]	Chen et al. [144], Jha et al. [145], Nakajima et al. [146]	
Educational AI & Learning Systems		Qiu et al. [147], Silva et al. [148]	Ho et al. [149], Wang et al. [150]	Shi et al. [151]

and found that the model frequently hallucinates references; a staggering 86% of the citations provided by the model did not actually exist. Similarly, Moayeri et al. [33] developed WorldBench to probe the factual recall of 20 state-of-the-art LLMs, discovering significant geographic disparities in their performance. Their analysis revealed that error rates for factual questions about countries in Sub-Saharan Africa were 1.5 times higher than for those in North America, highlighting systemic biases in the models' internal knowledge bases. Other works in this category extend the analysis to different

facets of model behavior and modality. Kraft et al. [65] investigated whether internal model enhancements, specifically knowledge enhancement, can inherently reduce bias. Their empirical evaluation demonstrated that such augmentations do not necessarily prevent stereotypical associations, suggesting that, without external corrective systems, even enhanced models can perpetuate epistemic injustice. In the multimodal domain, Jiang et al. [77] analyzed the representation space of an unaugmented LLM, identifying a significant gap between textual and visual representations as a key factor

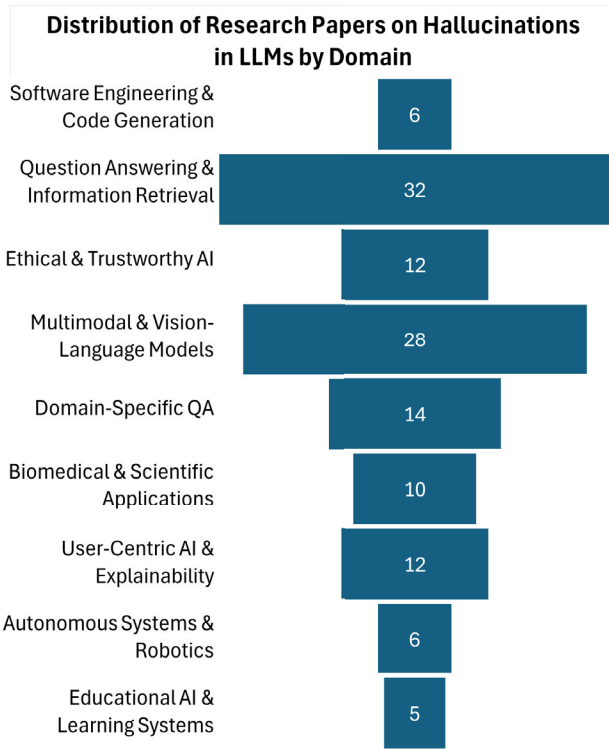


FIGURE 6. Distribution of the research papers across the nine identified domains.

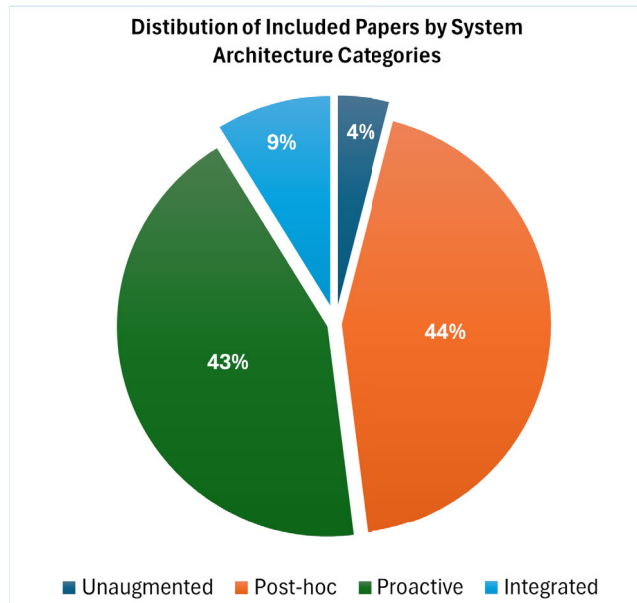


FIGURE 7. Distribution of the research papers across the four system architecture categories.

contributing to hallucinations. This foundational analysis motivated their subsequent development of a new training methodology.

Finally, Kaul et al. [78] introduced the THRONE benchmark to assess hallucinations in open-ended, free-form

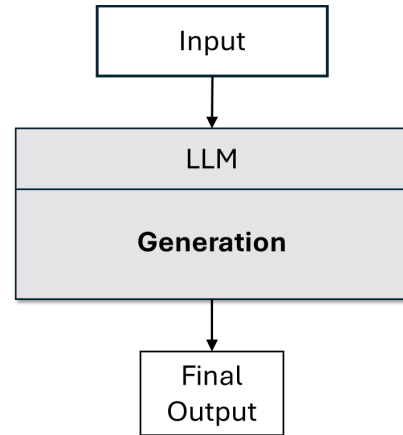


FIGURE 8. Conceptual illustration of an unaugmented LLM generation architecture.

responses. Their work reveals that mitigating hallucinations in more constrained settings (e.g., yes/no questions) does not correlate with a reduction in these more spontaneous, unprompted fabrications, underscoring the need for benchmarks that evaluate LLMs in their unconstrained, generative state.

Each of these studies examines models in the absence of external grounding or structured correction mechanisms, making them prototypical examples of Unaugmented LLM Generation Architectures as defined by this framework.

2) HIGHLIGHTING POST-HOC REACTIVE VALIDATION ARCHITECTURES

Post-hoc reactive validation architectures address LLM hallucinations by employing standalone detection and mitigation systems that operate after the LLM has generated its output, as illustrated in Figure 9. These systems typically react to outputs based on confidence thresholds [130], heuristics [35], [36], [89], or anomaly signals [45], [67], with corrections occurring externally without modifying the core LLM. Evaluation often emphasizes the precision and recall of detection.

A common strategy across various domains involves RAG to ground LLM outputs in external knowledge. In Question Answering & Information Retrieval, ReID is proposed as a robust hallucination detection discriminator trained on the RelQA dataset, leveraging ELECTRA as a backbone to align with human ratings [35]. FaCTQA detects and localizes factual errors in summaries by generating QA pairs from keywords and semantically matching answers from the summary and source text. It outperforms baselines, achieving 90.3% accuracy and 0.8701 F1 Score on XSum [36]. Similarly, in Domain-Specific QA, Complaint-LLM uses a knowledge-graph-driven approach with a KGCN classifier to enhance factuality in complaint processing, achieving peak accuracy/F1 scores of 0.85/0.86 [105]. ReRag optimizes vector database hyperparameters in a closed feedback loop, improving average context similarity scores from 0.494

(before optimization) to 0.616 (after optimization) [40]. For multimodal models, RAG combined with a Visual Hallucination Corrector in MM-PEAR-CoT enhances Visual Spatial Description tasks, improving SPICE scores to 86.61 [86]. NewsGPT, a 70B-parameter LLM, integrates a RAG module for hallucination control in news generation, achieving top performance in sentence-level discrimination (52.4% accuracy) [110].

Another approach focuses on leveraging internal model artifacts or self-assessment mechanisms for detection. In Software Engineering & Code Generation, empirical evaluation shows ChatGPT's self-verification is unreliable, with direct question prompts yielding high precision but very low recall (e.g., 0.02 for code generation). Guiding question prompts improve recall (+69% in vulnerability detection) but increase false positives. Self-contradictory hallucinations are frequently observed, especially in complex languages like Scala [27]. For general LLMs, uncertainty estimation techniques are explored; sample-based methods like Sample VRO show strong negative correlations (up to -0.739) with erroneous outputs, aiding in error detection without internal model access [69]. Snyder et al. [45] demonstrate that early detection of factual hallucinations is possible by analyzing internal model artifacts like self-attention and FC activations, achieving AUROC > 0.80 . Huo et al. [39] explore LLM self-detection of hallucinations in open-domain QA using retrieval-based techniques, finding that post-generation evidence retrieval can identify and correct hallucinations with moderate success.

Furthermore, several frameworks integrate rule-based or logic-based validation. Drowzee, a logic-programming-aided metamorphic testing framework, detects fact-conflicting hallucinations by deriving new facts through logic rules and comparing LLM output graphs to ground-truth graphs. Hallucination rates ranged from 24.7% (GPT-4) to 59.8% (Mistral-7B) [71]. For embodied agents in Autonomous Systems & Robotics, a two-stage alignment framework mitigates action hallucinations by combining parameter-efficient fine-tuning with RAG, achieving 100% Language Compliance [142]. CoT-TL translates natural language planning instructions into Linear Temporal Logic (LTL) formulas, integrating a model checker to reject invalid LTL formulas and self-consistency for robustness, achieving up to 91.69% accuracy [143]. In Biomedical & Scientific Applications, IC-BERT, a BERT-based instruction classifier, pre-screens user input to reduce hallucinations in TCM queries, achieving 99.95% accuracy [119]. User-centric AI initiatives like HILL aim to help users detect hallucinations through interface features like confidence scores and source links, showing increased hallucination awareness (+83.4%) in user studies [130].

In each case, the LLM remains a “static, unrefined generator”; validation is delegated to a standalone module that fires reactively and either blocks unsafe content or flags and corrects it after the fact. The lack of in-flight reasoning correction, coupled with an exclusive focus on

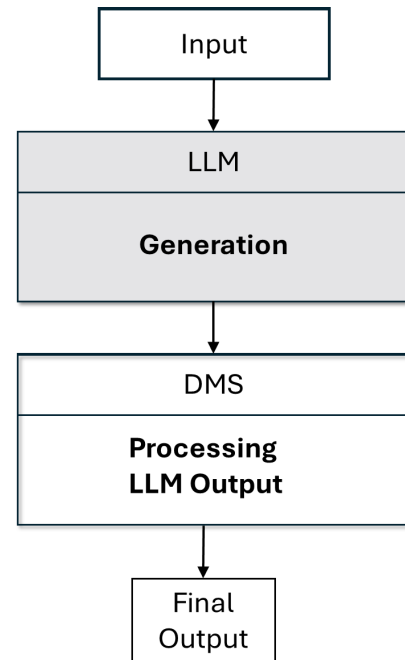


FIGURE 9. Conceptual illustration of a Post-hoc reactive validation architecture.

precision/recall of these downstream detectors, is precisely why this work belongs in the subsection “Post-hoc Reactive Validation Architectures”.

3) HIGHLIGHTING PROACTIVE DMS ARCHITECTURES

Research in this category shifts from post-hoc validation to proactive intervention, where the DMS is tightly coupled with the LLM to guide and constrain the generation process in real time. These architectures are defined by their ability to anticipate and preemptively correct errors, often through multi-stage pipelines, formal verification, and iterative feedback loops.

A dominant approach involves enhancing RAG with verification stages. In Question Answering, frameworks decompose complex queries into a series of sub-questions, where each reasoning step must be grounded in retrieved evidence before the final answer is synthesized [53], [56]. Some systems embed fact-checking directly into the retriever pipeline, pruning irrelevant or non-factual documents before they can influence the generator [48], or use real-time analysis of token entropy to trigger retrieval only at points of high uncertainty, improving efficiency [57]. This principle extends to domain-specific applications, where structured data from financial reports or manufacturing glossaries is converted into hierarchical chunks, and queries are enhanced with verified terminologies before LLM inference [111], [112], [114].

In high-stakes domains like Software Engineering & Code Generation, proactive systems integrate formal methods to enforce correctness. For instance, Refine4LLM uses a theorem prover to formally verify each refinement step suggested

by the LLM, creating a robust generation-validation loop that ensures code is provably correct [29]. Similarly, other frameworks employ multi-phase filtration pipelines that guarantee generated unit tests successfully build, pass reliably, and increase code coverage. At Meta, TestGen-LLM achieved a 75% build success rate, 57% pass rate, and 25% of the test cases increased coverage, significantly reducing hallucinations through formal guarantees [28]. This verification-in-the-loop pattern is also applied in Autonomous Systems, where formal verifiers check the LLM's plans and feed counterexamples back into the prompt for iterative correction. Preliminary experiments show that this approach can successfully eliminate identified hallucinations through iterative refinement using counterexamples, thereby converging toward verified outputs [145].

Multimodal systems employ novel decoding-time strategies that actively reshape the generation path. Techniques like DOPRA and OPERA monitor internal self-attention maps during generation, penalizing or rolling back the process when patterns indicative of “over-trust” in summary tokens emerge, thereby preventing the model from deviating from visual facts [95], [100]. Other frameworks use fine-grained correctional human feedback on specific hallucinated segments to retrain the model with a loss function that heavily prioritizes corrected regions. For instance, RLHF-V reduced the object hallucination rate by 34.8%, surpassing baselines trained with $7\times$ more data [76]. Despite these advances, failure modes persist; for example, formal verifiers can time out on complex specifications (Scenario 5), and RAG systems can be corrupted by noisy or outdated retrieved data (Scenario 7), highlighting that even proactive systems require careful validation of the DMS itself [29], [116].

Ultimately, these architectures signify a fundamental shift, treating the LLM not as a static generator to be checked but as an active component in a dynamic reasoning and verification loop. The emphasis on in-flight correction and preemptive guidance is precisely what defines these systems as proactive.

4) HIGHLIGHTING INTEGRATED LLM-DMS ARCHITECTURES

Integrated LLM-DMS architectures, as illustrated in Figure 11, mark the current frontier in hallucination mitigation by dissolving the traditional boundary between generator and detector. In these systems, the DMS is woven directly into the reasoning loop of the language model and uses internal feedback, meta-reasoning, and self-validation to anticipate, intercept, and correct errors before they occur.

Early examples include the retrieval-optimization-validation chain of Shi et al. [62], in which a self-critique module continuously refines the retriever's reasoning chain, and FS-C of Sun et al. [63], which enriches this loop with multi-perspective filtering to block faulty evidence in flight. The CORE framework by Wang et al. [64] adds causal graphs and Pearl-style interventions so that scoring and interventional regeneration jointly increase factual reliability. Experiments showed that CORE improved accuracy on

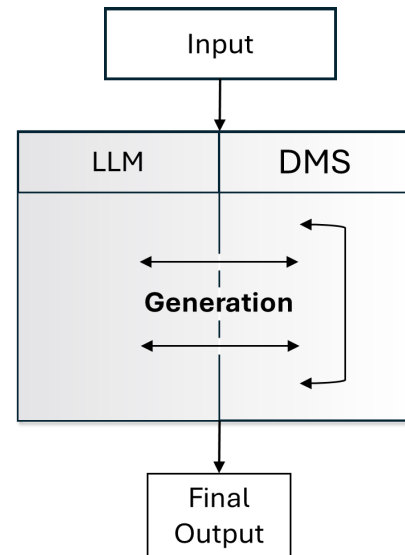


FIGURE 10. Conceptual illustration of proactive DMS architectures.

HotpotQA by 8.4% and reduced hallucination by 11.3% on ScienceQA compared to state-of-the-art baselines.

Other work focuses on linking intermediate inferences to explicit memory structures. The FACT framework by Gao et al. [102] couples rationale generation with post-hoc faithfulness checks and feeds validated rationales back into inference, while the GROUNDHOG system by Zhang et al. [104] ties language reasoning to symbolic visual memory so that a holistic reasoner can critique and revise multimodal outputs. GROUNDHOG achieves 22.6% lower object hallucination rates on Grounded Image Captioning (GIC) tasks and outperforms prior state-of-the-art methods without any task-specific fine-tuning. Complementing these architectures, the LVLM-eHub benchmark by Xu et al. [103] provides systematic, human-annotated probes for object hallucinations, supporting the co-development of integrated vision-language models.

Domain-specific integrations bring additional guarantees. In combustion science, Sharma and Raman [127] layer reference verification, physics-guided checks, and structured fallback generation onto a RAG backbone, maintaining scientific fidelity across literature review and simulation planning. MedLaSA [128] applies causal tracing to edit only the neurons responsible for a faulty fact, improving medical accuracy without disrupting unrelated knowledge. Sun et al. [139] their adaptive in-context learning paradigm and Zhang et al. [140] their empathetic conversational recommender both add auto-feedback and sentiment-aware reweighting to keep personalised outputs consistent with user intent and emotion, whereas Shi et al. [151] their multi-agent collaborative filtering framework distributes validation across interacting agents, reducing residual errors in the most complex Scenario 5-7 settings. Specifically, the MCF method improved average accuracy across ten datasets by 9.1% over

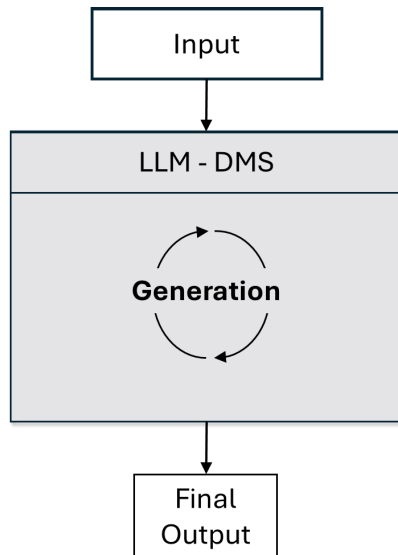


FIGURE 11. Conceptual illustration of an integrated LLM-DMS architecture.

baseline and achieved 75% valid reasoning trajectories in symbolic tasks.

The defining feature of these studies is a closed, anticipatory control loop. The DMS interrogates intermediate reasoning, intervenes before an error is finalized, and self-improves when its own judgments prove unreliable. By unifying generation, detection, and correction in a single dynamic pipeline, integrated LLM-DMS architectures set a new benchmark for long-term factual robustness.

V. DISCUSSION

A. KEY CHALLENGES AND OPEN ISSUES

One notable finding of our study is the uneven distribution of domains in hallucination research. Two areas - Question Answering & Information Retrieval and Multimodal & Vision-Language Models - account for 48% of all contributions, while the remaining seven areas account for only 52%, indicating a strong bias. This is particularly noticeable in areas such as Software Engineering & Code Generation (only 6 contributions, 4.8%), which is surprising given the widespread promotion of LLMs as a tool for programming tasks. Also underrepresented are the areas of Educational AI & Learning Systems, as well as Autonomous Systems & Robotics, which are only researched to a limited extent despite their growing social importance.

The discrepancy is even clearer when looking at the architectural sophistication of the systems. No contribution from the area of Software Engineering & Code Generation was rated above the Proactive DMS Architectures category, whereas in the areas of Question Answering & Information Retrieval [62], [64], [139] as well as in the areas of Multimodal & Visual-Language Models [102], [103], [104], three contributions each exceeded this threshold and thus demonstrated more advanced and integrated approaches to detecting and mitigating hallucinations.

In high-stakes domains like Biomedical & Scientific Applications, the need for hallucination control is critical. Encouragingly, research by Sharma and Raman [127] and Xu et al. [128] presents techniques that surpass traditional reactive correction models, indicating the development of robust frameworks that significantly reduce hallucinations, even extending to fully Integrated LLM-DMS Architectures.

On the other hand, domains such as Ethical & Trustworthy AI, while not underrepresented in terms of quantity, show no contributions rated at the Integrated LLM-DMS Architectures category. Although two notable efforts [75], [76] achieved Proactive DMS Architectures, most papers remain at a Post-hoc Reactive Validation Architectures level, indicating an opportunity for improvement in implementing more advanced and integrated mitigation strategies.

Other areas, such as Domain-Specific QA, show a high level of architectural sophistication in many papers (eight papers rated Proactive DMS Architectures), but still lack Integrated LLM-DMS Architectures, revealing a ripe opportunity for further advances. Similarly, Autonomous Systems & Robotics are not only underrepresented, but also do not show a mature application of advanced architectural frameworks, highlighting the need for targeted future research.

B. THEORETICAL IMPLICATIONS

The analysis of techniques related to hallucinations based on system architecture has shown that many current approaches remain heuristic or dominated by single-stage processing, particularly in less-researched areas. The utility of the framework lies in highlighting which domains have evolved towards deliberative, evaluative, and multi-phase processes (Proactive DMS Architectures and Integrated LLM-DMS Architectures) and which continue to rely on reactive or single-layer solutions.

The fact that only 11 papers (8.8%) achieved Integrated LLM-DMS Architectures reflects a greater challenge in aligning machine reasoning with highly integrated and sophisticated architectural designs. It raises the question of whether more deeply integrated architectural modeling is needed to advance hallucination mitigation beyond superficial corrections.

Moreover, success in high-stakes domains such as Biomedicine & Scientific Applications suggests that risk awareness can drive architectural sophistication in system development. This insight can help theorists develop context-sensitive architectural extensions that are better suited for domain-specific challenges.

C. PRACTICAL IMPLICATIONS

From a practical perspective, this review provides a diagnostic overview of where hallucination research is concentrated, and where it is not. Practitioners working in underrepresented domains such as Software Engineering & Code Generation, Educational AI & Learning Systems, and Autonomous Systems & Robotics should be aware of the methodological gaps and consider adapting techniques from more architecturally

sophisticated domains. The lack of Integrated LLM-DMS Architectures in these areas suggests that current systems may not be adequately equipped to detect and/or prevent hallucinations in high-impact scenarios.

In contrast, the architectural sophistication seen in Biomedical & Scientific Applications domain shows that multi-phase, knowledge-grounded, and uncertainty-aware architectures are both feasible and beneficial [127], [128]. Developers and product teams may draw from these domains to implement cross-domain mitigation techniques or to design hybrid models that combine strengths across research areas.

Finally, the taxonomy and mapping presented in this paper serve as a reference point for both researchers and engineers. It provides a foundation for benchmarking future techniques, understanding their architectural characteristics and integration levels, and identifying transferable practices.

D. CROSS-DOMAIN TRANSFERABILITY OF INTEGRATED LLM-DMS ARCHITECTURES

If we assume that Integrated LLM-DMS Architectures represent a real advance in mitigating hallucinations by enabling systems to anticipate, recognize, and correct errors through reasoning and meta-reasoning mechanisms, then a crucial next question arises: *Can these advanced architectural approaches be meaningfully adapted to domains beyond those for which they were originally developed?*

Empirical support for such cross-domain portability is growing. For instance, in the ontology-engineering study by Doumanas et al. [14], a four-level human-to-LLM collaboration scheme was applied to both medical and search-and-rescue ontologies, demonstrating that staged, human-in-the-loop designs remain effective when ported across knowledge domains. Further evidence comes from research on high-fidelity training datasets. Dziri et al. [152] created FAITHDIAL, a benchmark of hallucination-free dialogues, and found that models trained on this curated data showed remarkable generalization. In a zero-shot transfer setting, their model significantly reduced hallucinations when tested on other knowledge-grounded dialogue datasets like TopicalChat and CMU-DoG, proving more faithful than models trained specifically on the in-domain data. Similarly, the successful application of general-purpose architectures to highly specialized fields highlights this transferability. Lee et al. [153] demonstrated that a RAG model, a common strategy for mitigating hallucination, outperformed general-purpose models like GPT-4 in the specialized domain of construction safety management. Their work underscores how architectural solutions designed for broad knowledge-intensive tasks can be effectively adapted to improve factual grounding and reliability in niche, high-stakes domains.

While approaches categorized as Integrated LLM-DMS Architectures show promising results in domains such as Biomedicine, Multimodal AI, and Educational Systems,

their architectures often remain tightly coupled to their domain-specific assumptions. This raises the question of whether and how they can be transferred to underrepresented yet practically important areas such as Software Engineering & Code Generation, Ethical & Trustworthy AI, Domain-Specific Questioning & Answering, and Autonomous Systems & Robotics and what adaptations, limitations, or risks might be involved in doing so.

A comparative review of the Integrated LLM-DMS Architecture contributions reveals several architectural characteristics that are not only advanced but also potentially transferable across domains. Notable examples include multi-agent validation mechanisms [151], causal knowledge pipelines [64], adapter-based knowledge editing with minimal collateral effect [128], and retrieval-augmented generation with embedded citation verification [127]. These strategies often exhibit modularity, feedback loops, or self-correction mechanisms that can be adapted with reasonable effort.

Other approaches, such as rationale distillation through symbolic execution and natural language transformation [102], sentence-weighted reasoning validation [63], or grounding via segmentation and pointer tokens [104], also reveal underlying principles that could be tailored for different domain challenges. Although frameworks like LVLM-eHub [103] serve primarily as benchmarks rather than intervention mechanisms and are therefore excluded from transferability, the vast majority of Integrated LLM-DMS Architecture contributions indicate opportunities for reuse in different system environments.

By analyzing the papers classified as Integrated LLM-DMS Architectures in conjunction with the defined application domains, it was possible to identify components and architectural principles within these approaches that hold strong potential for cross-domain transfer. Specifically, the analysis revealed that several of these methods exhibit modular reasoning units, meta-validation layers, or causal grounding strategies that could be meaningfully adapted to domains that are currently underrepresented in the literature. The following discussion explores this transferability along four underrepresented domains: Software Engineering & Code Generation, Ethical & Trustworthy AI, Domain-Specific Question Answering, and Autonomous Systems & Robotics, highlighting the rationale for adaptation, possible benefits, and domain-specific constraints.

1) SOFTWARE ENGINEERING & CODE GENERATION

Several Integrated LLM-DMS Architecture approaches show strong promise for transfer into the Software Engineering domain. For instance, the Fact framework [102], which generates verifiable and concise rationales via symbolic execution and natural language transformation, could significantly benefit automated code explanation, debugging, and unit test generation. The symbolic-to-natural logic transformation pipeline aligns with the needs of explainable and deterministic software behavior modeling.

Similarly, Sharma and Raman [127] citation-verified RAG pipeline, initially designed for combustion science, could be adapted to structure large-scale code repositories and deliver grounded, citation-supported answers in software maintenance or DevOps scenarios.

2) ETHICAL & TRUSTWORTHY AI

Several frameworks, including Multi-agent Collaborative Filtering (MCF) [151] and CORE [64], offer mechanisms that could improve trust, explainability, and ethical compliance in generative systems. For example, MCF's multi-agent verification and AMR-based alignment may help in validating ethical consistency across diverse outputs. Likewise, CORE's causal reasoning pipeline could be employed to trace and justify ethically sensitive decisions.

MedLaSA [128], while tailored to biomedical knowledge, also provides fine-grained editing and control with minimal collateral effects, features vital for updating knowledge bases in high-stakes ethical scenarios such as regulatory changes or bias mitigation.

3) DOMAIN-SPECIFIC QA

The causal pipeline in CORE [64], with its use of belief scoring and do-calculus, is directly relevant to structured QA tasks in domains such as law, manufacturing, or finance. Likewise, MedLaSA's [128] scalable adapter injection could allow fine-grained factual corrections in specialized corpora, such as technical standards or compliance regulations.

Furthermore, the Groundhog framework [104], while originally multimodal, offers a transparent grounding mechanism that could support visual QA tasks in technical drawings or diagnostic imaging.

4) AUTONOMOUS SYSTEMS & ROBOTICS

Transferability into autonomous systems requires high reasoning robustness, low latency, and explainable decision-making. In this context, GROUNDHOG's segmentation-based grounding and multi-modal validation [104] could help mitigate hallucinations in vision-based robotic control. Similarly, MCF's agent-based architecture [151] aligns well with ensemble-based decision-making in robotics.

Additionally, Fact's rationale validation mechanism [102] may aid in producing human-interpretable justifications for autonomous decisions, a key requirement for human-in-the-loop systems.

While these domain-specific transfer scenarios show promising ways to reuse the techniques of Integrated LLM-DMS Architectures, they are not without limitations. Cross-domain adaptation inherently involves challenges that must be addressed to ensure effectiveness and reliability. In resource-poor domains, the lack of adequate ground truth [64] could compromise the effectiveness of validation mechanisms. Uncalibrated adaptation could also lead to semantic drift, where correction mechanisms misrecognize or distort domain-specific reasoning patterns [63], [102]. Furthermore, the increasing modularity and abstraction of

Integrated LLM-DMS Architectures might benefit generalization but reduce transparency, leading to trust and explanation problems in high-stakes domains [102], [104].

Nevertheless, the reviewed approaches collectively represent a rich design space for hallucination mitigation, many of which hold potential beyond their initial context.

By fostering architecture transfer from mature to under-represented domains and modular adoption of Integrated LLM-DMS Architecture components, hallucination research can be extended to broader application domains, ultimately helping to close the gap between system reliability and real-world deployment.

E. BRIDGING LLM ARCHITECTURES WITH DUAL PROCESS THEORY

Beyond the architectural classification and transferability insights discussed in this review, future research could benefit from a deeper exploration of the cognitive underpinnings of hallucinations in LLMs. In particular, the application of Dual Process Theory (DPT) offers a promising complementary framework for understanding how LLMs generate outputs, especially in contexts prone to factual inaccuracy.

DPT, rooted in foundational work from cognitive psychology and decision science [154], [155], [156], [157], distinguishes between two qualitatively different cognitive modes: System 1, which is fast, intuitive, and associative, and System 2, which is slow, reflective, and rule-based. While this review has focused primarily on system-level architectural classifications of hallucination mitigation strategies, future work could explore whether and how these DPT constructs can be operationalized within the internal mechanisms of LLMs.

Recent advances indicate that such integration is feasible. For instance, Cheng et al. [158] introduce HaluSearch, a framework that employs tree-search-based slow thinking and a dynamic System 1 / System 2 switch to mitigate hallucinations during inference. This approach maps DPT concepts onto computational processes, replacing metaphor with empirically testable mechanisms such as step-level, self-evaluation and hierarchical reasoning selection. Such implementations provide a promising direction for further empirical investigation.

Nonetheless, existing DPT-inspired models for LLMs have been critiqued for oversimplification or ad hoc conceptual mapping [159]. To move beyond metaphorical alignment, future research should aim to establish concrete empirical mappings between neural activations or internal states in LLMs and cognitive constructs from DPT. For example, studies could investigate whether "slow thinking" modules, such as meta-cognitive loops or deliberative planning agents, correlate with measurable improvements in factual consistency, response calibration, or self-correction behaviors.

A more granular application of DPT may also illuminate failure modes in current systems. Whereas most current hallucination mitigation strategies either operate globally

(e.g., via training data quality) or post hoc (e.g., through faithfulness scoring), a DPT-inspired approach could enable real-time cognitive modulation of reasoning depth based on context complexity. This opens up the possibility of adaptive control mechanisms that switch between System 1-like generation and System 2-like reasoning based on task demands.

Finally, future work should also investigate the boundary conditions and limitations of DPT when applied to artificial systems. As Evans [155] emphasizes, dual-process theories encompass diverse and sometimes conflicting accounts, and not all DPT attributes may sensibly map onto LLM architectures. Moreover, as Norman et al. [157] and others note, both intuitive and deliberative processes can contribute to error under specific conditions. A critical research priority will thus be to empirically identify when and how slow-thinking mechanisms meaningfully improve reliability without introducing unacceptable computational overhead or unintended consequences.

In sum, by systematically investigating the relevance and implementation of DPT-inspired constructs in LLM architectures, future research can move beyond superficial analogies and toward principled, theory-guided system design. This may ultimately lead to the development of more robust, explainable, and human-aligned generative models.

VI. CONCLUSION

This paper presents a SLR of hallucinations in LLMs, which provides a domain-based classification integrated into a descriptive organizational scheme based on system architectures. By analyzing 125 peer-reviewed publications, we were able to identify patterns in the way hallucination detection and mitigation techniques are applied in nine application domains and how they relate to different levels of architectural sophistication and integration.

Our findings show that research efforts are concentrated in the areas of Question Answering & Information Retrieval, as well as Multimodal & Visual-Language Models, which together account for almost half of all studies (48%). These domains also exhibit higher architectural sophistication, including several approaches that demonstrate Integrated LLM-DMS Architectures by incorporating multiphase processing, knowledge-based, and uncertainty-aware components. In contrast, areas such as Software Engineering & Code Generation, Educational AI & Learning Systems, and Autonomous Systems & Robotics are underrepresented and underdeveloped despite their practical relevance in terms of hallucination resilience.

A. LIMITATIONS

While this review provides a comprehensive account of techniques related to hallucinations, several limitations must be acknowledged.

First, the inclusion criteria and keyword filters may have inadvertently excluded relevant papers, particularly those that used non-standardized terminology or focused on adjacent

tasks (e.g., summarizing or paraphrasing) without explicitly identifying them as hallucination-related.

Second, although guided by a standardized rubric, the assignment of system architecture categories involves a certain degree of subjectivity, especially for borderline cases or hybrid methods that do not clearly fit into the defined categories. In addition, the classification of domains may be fluid for some interdisciplinary work, which could easily influence the number of domains.

Third, a further limitation concerns the organizational scheme applied in this review. Specifically, we employed a set of simplified architectural categories to classify detection and mitigation strategies. While these abstractions provided a structured and comprehensible lens through which to evaluate the literature, they also simplified the rich and often overlapping features of complex LLM-DMS interactions. By mapping hallucination-related techniques in LLMs onto these simplified categories, we risked compressing the underlying algorithmic dynamics into discrete classifications that may not fully capture their nuances. However, adopting this simplified framework enabled a systematic comparison across a large body of work and supported the creation of a unified classification scheme. Future work could extend this analysis using more granular or hybrid architectural models that reflect new insights into machine reasoning and provide a more in-depth explanation of how hallucinations arise and are processed in artificial systems.

Fourth, although several promising Integrated LLM-DMS Architecture approaches with cross-domain potential have been identified in this review, the actual process of transferability remains largely theoretical. We have not experimentally tested how well these frameworks work outside their original domains. The effectiveness, scalability, and potential limitations of these techniques in adapting to new domains, therefore, remain an open empirical question that warrants future investigation.

Lastly, our analysis does not assess quantitative performance metrics for the different papers, but focuses on methodological features, which may limit direct comparability in terms of effectiveness.

B. FUTURE RESEARCH

This work presents a theoretical classification of hallucination mitigation techniques using a domain-based framework aligned with system architectures. Building on the results of this study, several areas can be explored in future research.

One important direction is the empirical validation and practical adaptation of Integrated LLM-DMS Architecture techniques in underrepresented domains. While this review already analyzed the transferability potential of these advanced architectural approaches across domains such as Software Engineering, Educational AI, and Autonomous Systems this analysis remains conceptual. What is now required are applied studies that benchmark these architectures in target domains, under real-world conditions. Techniques such as multi-agent validation [151], and causal reasoning

pipelines [150] have shown promise in a cursory analysis in Ethical & Trustworthy AI and Domain-Specific QA domains. Their adaptation must now be tested empirically to assess not only effectiveness but also implementation feasibility, performance trade-offs, and integration challenges within new domain-specific workflows.

To support this, future work should also aim to isolate the most impactful architectural modules through ablation studies and module-level benchmarking. Evaluating components such as feedback loops, meta-reasoning layers [62], and retrieval-integrated validators [127] could guide the design of modular toolkits for cross-domain reuse. Bridging the gap between theory and practice will require not only empirical validation but also the systematic development of reusable and adaptable architectural building blocks.

Additionally, the subjective nature of architectural sophistication assessment and the variability in domain boundaries present challenges that may be addressed in future studies by developing more fine-grained evaluation rubrics or automated classification tools. This could also include the creation of a modular benchmarking framework that systematically compares components such as retrieval integration, error correction, and feedback loops across tasks.

Findings from the discussion also suggest potential in exploring hallucination behaviors in multimodal and conversational systems, where factors like co-reference resolution and modality fusion can introduce new types of hallucination [93]. Finally, deeper integration of user feedback and system alignment mechanisms, including active learning, reinforcement learning, or interactive error correction, may help advance the architectural sophistication of techniques and improve the robustness of LLM-based systems.

In summary, future research has the opportunity not only to close domain-specific gaps but also to advance theoretical models toward a more unified, empirical, and context-sensitive understanding of hallucination in LLMs.

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